Water Supply and Wastewater Engineering

CEP REPORT

SEWERAGE SYSTEM



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2022-EN-21

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1 Executive Summary

The purpose of this report is to design an efficient and sustainable sewerage system for the Muslim Nagar Housing Society, a growing residential area in Lahore that currently relies on outdated wastewater management practices. This initiative aims to address critical issues like water stagnation, health risks, and environmental hazards, while accommodating the community's future needs.

The study highlights the area’s lack of a centralized sewerage system, uneven terrain, and reliance on septic tanks and informal drainage. It proposes a partially combined sewerage network incorporating a gravity-based system supplemented by a pumping station to handle topographical challenges. Advanced modeling tools like SewerGEMS were employed to optimize the network's layout, ensuring compliance with self-cleansing velocity standards and PHED guidelines.

Key design features include:

1. Gravity-driven sewage flow for efficiency and low operational costs.
2. RCC pipes of optimized diameters to handle peak flows and resist external loads.
3. Strategically placed manholes for maintenance and flow regulation.
4. A wet well and a pumping station with standby pumps for reliability.

The hydraulic and structural designs, supported by comprehensive load analysis, validate the system's capacity to handle peak wastewater flow. Estimated capital costs total PKR 5.62 million, while annual O&M expenses amount to PKR 7.21 million. A per-household sewer bill of PKR 1,445/month ensures financial sustainability.

In conclusion, the proposed design offers a long-term solution that resolves current inefficiencies, meets future demands, and enhances public health and environmental quality. The report recommends timely billing and transparent fund utilization for smooth implementation and operational success.

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Abbreviations used

Psi: Pounds per square inch

LPCD: Liters per capita per day

GIS: Geographic Information System

CAD: Computer-Aided Design

GPM: Gallons per minute

GI: Galvanized Iron

PHED: Public Health Engineering Department

MRS: Market Rate Schedule

RCC: Reinforced Cement Concrete

HGL: Hydraulic Grade Line

OHR: Overhead Reservoir

O&M: Operation and Maintenance

PRA: Punjab Revenue Authority

2 Background

Muslim Nagar Housing Society is a growing residential area located along Jati Umrah Road, Lahore. Covering about 45 acres (18 hectares), the society is home to around 4,550 people and includes amenities such as schools, mosques, parks, and shops. While the society is well-planned in many ways, it currently lacks a proper sewerage system. Residents depend on septic tanks and informal drainage, which are not only inconvenient but also harmful to the environment and public health.

The area's uneven terrain adds to the complexity of managing wastewater. Without an organized sewerage network, water stagnation and improper sewage disposal have become common problems. These issues highlight the urgent need for a modern, centralized sewerage system that is efficient, safe, and future-proof. By designing a system that aligns with the latest engineering standards and takes the society's unique needs into account, this project aims to provide a long-term solution for wastewater management.



Figure 1. map layout

3 Project Area Description

Muslim Nagar Housing Society is a well-planned residential area located along Jati Umrah Road, Lahore, Pakistan. The society spans approximately **18 hectares** *(Google Earth)* and serves middle and upper-middle-class residents. Key details of the project area include

Coordinates that are approximately **31.5204° N, 74.3587° E** *(Literature)*, with easy access via the main Lahore Ring Road. Lahore experiences hot summers (up to **45°C**), mild winters (as low as **5°C**), and receives an average annual rainfall of **~625 mm**, predominantly during the monsoon season.

3.1 Facilities

Residential Plots are about **416 residential plots** of varying sizes*. (Society head office)*

Community Amenities includes **one school**, **one mosque**, **five parks**, and two **commercial** establishments.

The project area reflects a growing urban environment with specific challenges related to topography, climate, and wastewater generation. The proposed sewerage system design aims to address these challenges while meeting the needs of the community efficiently and sustainably.

4 Situation Analysis of Sewerage

Currently, Muslim Nagar Housing Society lacks a proper sewerage system, and the existing methods for wastewater management are both outdated and problematic. Most residents rely on septic tanks or informal drainage systems, which come with several drawbacks.

The reliance on septic tanks leads to frequent maintenance issues and high costs for households. Moreover, improper drainage channels result in stagnant water, creating an unhygienic environment and becoming breeding grounds for mosquitoes. These issues also pose significant health risks, increasing the likelihood of waterborne diseases such as cholera and typhoid.

The uneven terrain further complicates matters, as water collects in low-lying areas, leading to localized flooding during rains. Without a centralized sewerage network, the society faces challenges in efficiently managing its growing wastewater needs, especially as the population continues to rise.

Addressing these concerns is vital. A well-planned sewerage system will not only resolve the existing problems but also ensure a cleaner, healthier environment for all residents. By incorporating modern technology and engineering best practices, the proposed system will provide a sustainable and reliable solution for the community.

5 Proposal

To address the pressing issues of wastewater management in Muslim Nagar Housing Society, a modern sewerage system is proposed. The design focuses on creating an efficient, sustainable solution that caters to the community's needs for the next 20 years.

The system will be a **partially combined sewerage system**, capable of managing domestic wastewater along with minor surface runoff during the rainy season. Key features of the proposal include:

- **Gravity-Based Network:** Most of the sewage will flow naturally through a carefully designed gravity network, minimizing energy usage and operational costs.
- **Integration of a Pumping Station:** To handle the area’s uneven terrain, a pumping station will be strategically located to manage low-lying areas effectively.
- **Modern Technology:** The design will incorporate advanced modeling tools like SewerGEMS to optimize network layout, pipe sizes, and flow dynamics.
- **Compliance with PHED Standards:** The system will adhere to design criteria,

including minimum velocities, pipe slopes, and manhole placements, ensuring efficiency and durability.

This proposal aims to resolve existing inefficiencies while laying the groundwork for a cleaner, healthier, and more resilient community. By addressing topographical challenges and future population growth, the system ensures both immediate and long-term benefits for residents.

6 Design Considerations

Building on the challenges and objectives identified in the **Background section**, the design considerations for the sewerage system of **Muslim Nagar Housing Society** focus on ensuring technical reliability, cost efficiency, and adherence to **WASA** and **PHED Punjab guidelines**.

6.1 Design Criteria

1. **Sewage Generation**:

- Sewage generation is assumed to be **80% of water consumption**.
- Infiltration and inflow (I&I) allowance: **10% of total sewage flow**.

2. **Velocity Standards**:

- Minimum velocity for self-cleansing: **2.5 ft/s**.
- Maximum velocity to avoid erosion: **10 ft/s**.

3. **Pipe Material and Sizes**:

- Pipe material: **RCC (Reinforced Cement Concrete)**.
- Minimum pipe diameter: **9 inches (225 mm)**.

4. **Manholes**:

Manholes must be installed:

- At every **300 ft** along straight sections.
- At changes in direction or slope.
- At junctions where two or more sewers meet.

5. **Pipe Cover**:

Minimum pipe cover: **3 ft** to protect against external loads.

6. **Peak Flow Factor**:

A **peak factor of 2.5** is applied to account for peak flow conditions.

7. **Hydraulic Design**:

   Manning's Equation is used for pipe design with:

   - Manning's coefficient, n=0.013 (for RCC pipes).

6.2 Design Parameters

Using the design criteria, the following parameters are calculated for **Muslim Nagar Housing Society**:

1. **Sewage Flow Calculation**:

- Daily sewage flow (80% of water demand):

$$Sewage\ Flow\ (ft^3/day) = Water\ Demand\ (ft^3/day) \times 0.8$$

$$Water\ Demand\ ((ft\text{^}3)/day) = 385.45\ m\text{^}3/day(from\ water\ supply\ report)\ \times 35.31 = 13{,}647\ (ft\text{^}3)/day$$

$$Avg\ Sewage\ Flow = 13{,}647 \times 0.8 = 10{,}918\ ft^3/day$$

Infiltration and inflow (I&I):

I&I = Sewage flow * 0.1 = 10,918 * 0.1 = 1092 ft^3/day

- **Peak flow** (using peak factor of 2.5):

$$Peak\ Flow\ ((ft\text{^}3)/day) = Avg\ Sewage\ Flow\ (ft^3/day) \times Peak\ Factor$$

$$Peak\ Flow = 12{,}010 \times 2.5 = 30{,}025\ ft^3/day$$

Peak flow (in ft^3/s):

$$Peak\ Flow\ ((ft\text{^}3)/s) = 30{,}025/24 \times 60 \times 60 \approx 0.35\ ft^3/s$$

Storm Flow = Peak sewage flow

- **Total Sewage Flow** (including I&I):

Total Sewage Flow= 1092 + 30025 + 30025 = 61,142 ft^3/day

= 0.71 ft^3/sec

2. Hydraulic Design

The sewerage network for **Muslim Nagar Housing Society** was hydraulically designed using **SewerGEMS software**, supported by manual calculations outlined in the **Hydraulic Statement**. The design ensures compliance with self-cleansing velocity requirements, minimum pipe sizes, and sewer slope standards.

6.2.1 Sewer Network Layout

The **Hydraulic Statement** defines the network layout, which includes:

- The total number of sewer sections.
- Sewer lengths, slopes, and diameters for each section.
- Design flow rates for each sewer.

This layout ensures that wastewater is efficiently conveyed to the disposal point while preventing sedimentation or overflow.

6.2.2 Hydraulic Calculations

The **Hydraulic Statement** provides detailed calculations for:

1. **Flow Velocities**:

   Velocities in all sewer sections were evaluated to ensure:

   - **Minimum Velocity (2.5 ft/s)**: Prevents sedimentation.
   - **Maximum Velocity (10 ft/s)**: Avoids pipe erosion.

2. **Sewer Slopes**:

   Slopes were calculated using **Manning's Equation**:

   $$V = \frac{1.49}{n} \cdot R^{\frac{2}{3}} \cdot S^{\frac{1}{2}}$$

   Where:

   - V = Velocity ft/sec
   - R = P/A: Hydraulic radius, derived from the cross-sectional area (A) and wetted perimeter (P).
   - S = slope of sewer
   - n = 0.013 for RCC pipe

3. **Flow Capacities**:

   Each sewer's capacity was confirmed to exceed the **peak flow** (30,025 ft^3/day or 0.35 ft^3/s), including allowances for infiltration and inflow.

6.3 Software Simulation

Using **SewerGEMS**, the sewerage network was simulated to validate:

1. **Hydraulic Gradeline (HGL)**: Ensures the flow remains within the pipe and does not surcharge.
2. **Pressure Conditions**: Confirms the network operates under gravity flow.
3. **Critical Nodes**: Identifies high-risk sections requiring additional design considerations.

6.4 Summary of Design Outcomes

The Hydraulic Statement concludes:

- All sewer sections meet **self-cleansing velocity requirements**.
- Slope adjustments were made in sections with marginal velocities to ensure uninterrupted flow.
- The pipe diameters, as summarized in the Hydraulic Statement, were optimized to balance cost and performance.

6.5 Manhole Design

The inclusion of **manholes** in the sewerage system ensures accessibility for maintenance, cleaning, and flow regulation. The design of manholes is guided by **PHED Punjab standards** and tailored to the network layout provided in the **Hydraulic Statement**.

6.5.1 Placement Criteria

Manholes are strategically placed at:

1. **Straight Sections**: At intervals of **300 ft** to facilitate inspection and maintenance.
2. **Changes in Direction**: To manage flow and reduce turbulence.
3. **Junctions**: Where two or more sewers meet.
4. **Changes in Diameter or Slope**: To regulate flow transitions.

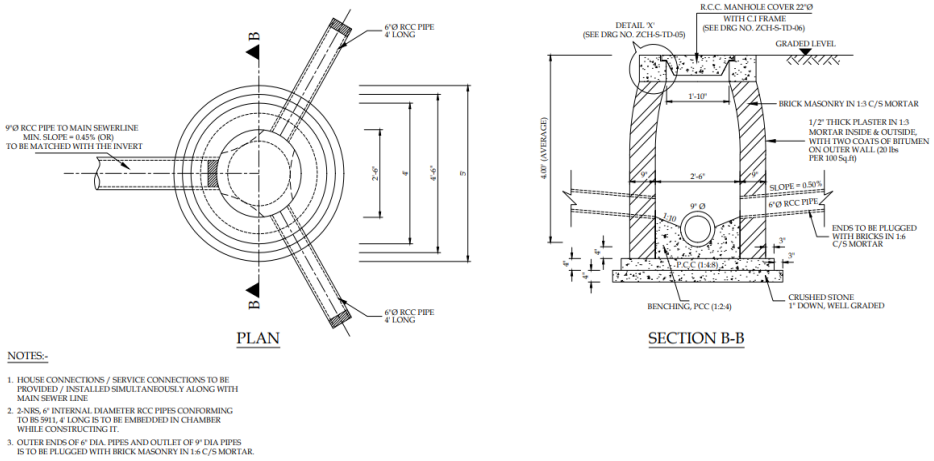


Figure 2. Plan and Cross Section View of a Manhole with depth up to 4'

6.5.2 Manhole Types

1. **Shallow Manholes**:
   - Depth: **<3 ft**.
   - Used at minor changes in slope or diameter.
2. **Intermediate Manholes**:
   - Depth: **3–6 ft**.
   - Commonly used in straight sections.
3. **Deep Manholes**:
   - Depth: **>6 ft**.
   - Used at junctions or where significant flow regulation is required.

6.5.3 Design Parameters

The **Hydraulic Statement** provides the data necessary for detailed manhole design, including:

1. **Total Number of Manholes**
2. **Manhole Dimensions**:
   - Internal diameter: **4 ft** for mainline sewers.
   - Cover: Designed to withstand traffic loads (minimum **3 ft thickness**).

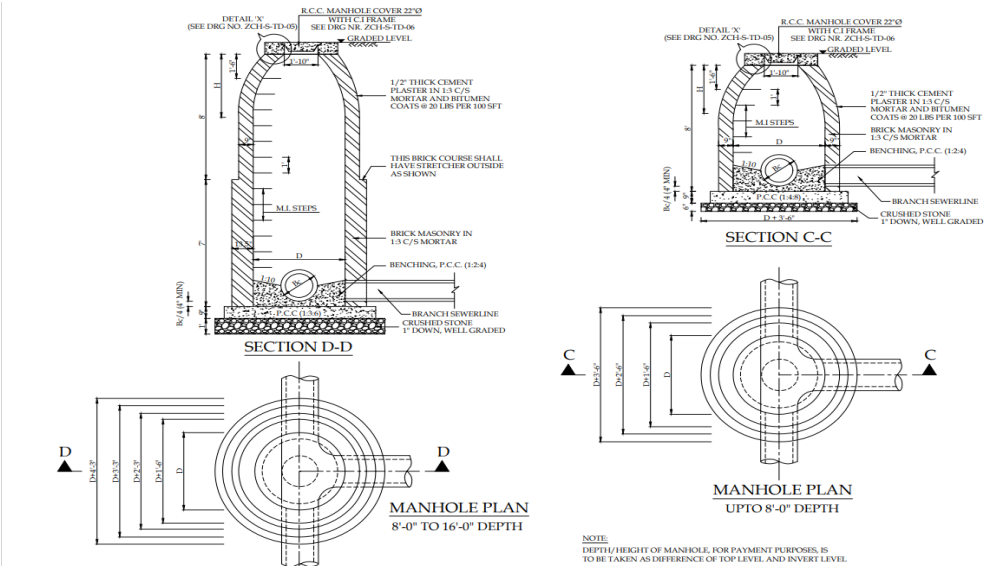


Figure 3. Sewer Pumping Station Manhole Plan Section D-D

6.5.4 Construction and Costs

1. **Materials**:

   - **Brick masonry** or **RCC** is used, depending on depth and location.
   - Heavy-duty covers for manholes are used in traffic areas.

2. **Estimated Cost**:

   Cost per manhole: PKR **20,000** for shallow manholes and PKR **50,000** for deep manholes (based on local market rates).

The manhole design ensures:

- Accessibility at critical points in the network.
- Proper regulation of flow, reducing risks of blockage or overflow.
- Compliance with **PHED Punjab standards** for safety and durability.

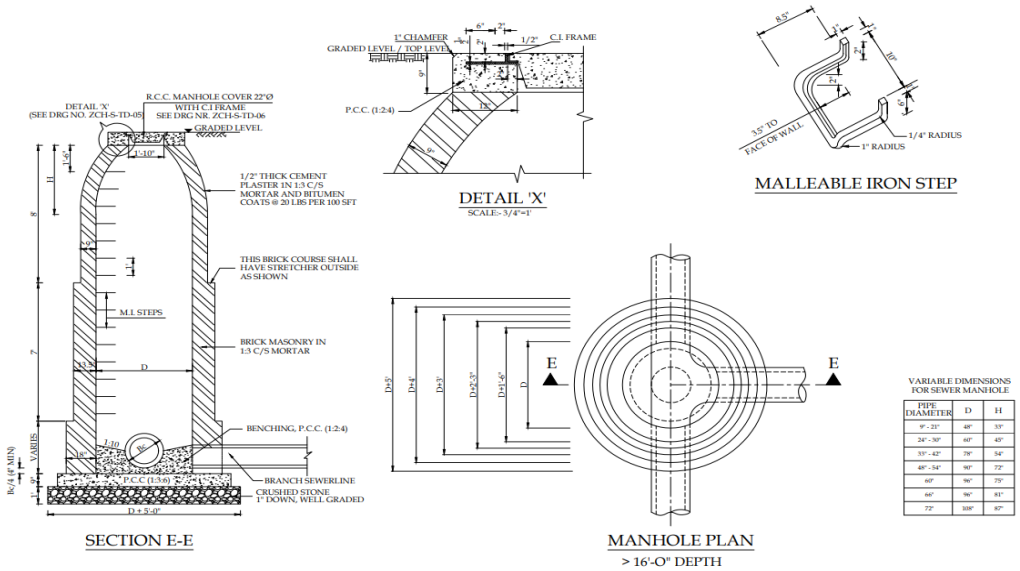


Figure 4. Sewer Pumping Station Manhole Plan Section E-E

7 Hydraulic Design (Continuation)

Building on the principles and software simulations discussed earlier, the sewerage system for **Muslim Nagar Housing Society** was hydraulically designed to ensure compliance with **PHED Punjab standards** and **self-cleansing velocity criteria**. Detailed hydraulic computations for all sewer lines were conducted to verify capacity, velocity, and slopes, as summarized below.

7.1 Hydraulic Computations for Sewer Lines

The sewerage network was divided into sections, and hydraulic calculations were performed for each using the data from the **SewerGEMS simulations**. The following parameters were considered:

1. **Sewer Diameter**: Optimized to balance cost and capacity.
2. **Sewer Length**: Taken from the network layout.
3. **Design Flow**: Includes peak flow and allowances for infiltration/inflow and stormwater.
4. **Velocity**: Ensures compliance with the minimum velocity requirement of **2.5 ft/s** and the maximum velocity limit of **10 ft/s**.
5. **Slope**: Derived using **Manning's Equation** to achieve the required velocity.

7.2 Hydraulic Computation Summary

The detailed hydraulic computations for all sewer sections, including diameters, slopes, velocities, and flow rates, have been thoroughly analyzed in the **Hydraulic Statement** attached in annexure. These computations verify that:

1. Not all sewer lines meet the minimum criteria of self-cleansing velocity and are below it, so

    - Sewer lines with velocity equal to or greater than 2 are considered OK
    - Sewer lines with velocity ranging from 1-2 ft/sec are given **Period Flushing**
    - Sewer lines with velocity ranging below 1 ft/sec are provided with **Regular Flushing**
    - Laterals have less than 1 ft/sec velocity so these are given flushing with **Flush Tanks**

2. Peak flow conditions are safely accommodated in each sewer section.
3. Critical sections have been designed to balance slope, diameter, and velocity.

8 Load Analysis on Sewer Pipes

The sewer pipes in **Muslim Nagar Housing Society** must be designed to withstand:

1. **Earth Loads**: The weight of the soil above the pipes.
2. **Traffic Loads**: Loads from vehicular movement on roads above the pipes.

To ensure structural safety, the following considerations are included in the design:

8.1 Earth Load Analysis

Methodology:
The earth load on the sewer pipes is calculated using Marston's formula:

$$W = C \cdot w \cdot B^2$$

Where:

- W = Earth load per unit length of the pipe (lbs/ft).
- C: Load coefficient, depending on soil type and trench conditions.
- w : Unit weight of soil (120 lbs/ft3 for clayey soil).
- B: Trench width (ft).

Assumptions for Design:

- Soil type: Clayey.
- Trench width: 2.5 × pipe diameter.

Calculations

- Soil unit weight: **120 lbs/ft³ (clayey soil)**
- Load coefficient (C): **0.12**
- Trench width (B): Calculated as 2.5 times the pipe diameter.

Table 1. Earth Load

Pipe Diameter (in)	Trench Width (ft)	Earth Load (lbs/ft)
9	1.88	50.62
12	2.50	90.00
15	3.13	140.63
18	3.75	202.50

8.2 Traffic Load Analysis

Standard Applied:
HS-20 vehicular loading (per AASHTO standards) is used to calculate traffic loads.
The load is distributed based on the depth of pipe cover:

$$Pt = \frac{P}{\left(1 + \frac{2H}{B}\right)^2}$$

Where:

- Pt : Traffic load on the pipe (lbs/ft²).
- P: Wheel load (16,000 lbs for HS-20).
- H: Depth of pipe cover (ft).
- B: Trench width (ft).

Calculations

- Wheel load (P): **16,000 lbs**
- Depth of cover (H): **3 ft**
- Trench width (B): Based on pipe diameter.

Table 2. Traffic load

Pipe Diameter (in)	Trench Width (ft)	Traffic Load (lbs/ft²)
9	1.88	907.03
12	2.50	1384.08
15	3.13	1876.52
18	3.75	2366.86

8.3 Total Loads on Pipes

The total load is the sum of earth and traffic loads:

Table 3. Total load

Pipe Diameter (in)	Earth Load (lbs/ft)	Traffic Load (lbs/ft²)	Total Load (lbs/ft)
9	50.62	907.03	957.65
12	90.00	1384.08	1474.08
15	140.63	1876.52	2017.15
18	202.50	2366.86	2569.36

8.4 Observations and Recommendations

8.4.1 Pipe Strength:

- To verify pipe safety, the total load (lbs/ft) must be compared to the D-load capacity of the selected pipe material (e.g., RCC pipes).
- Typically, RCC pipes can handle **D-loads ranging from 1,000–3,000 lbs/ft/ft** depending on wall thickness and reinforcement.

8.4.2 Proposed Bedding:

- For **9-12 inch pipes**: Granular bedding (minimum 6 inches) is sufficient.
- For **15-18 inch pipes**: RCC bedding is recommended in high-load areas.

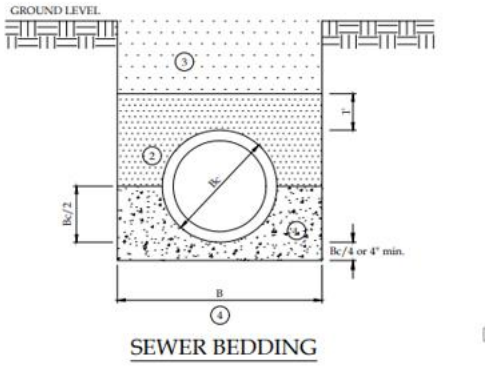


Figure 5. Sewer bedding

9 Pumping Station Design

For areas where gravity flow is insufficient or where wastewater needs to be lifted to a higher level, a **pumping station** is essential. In **Muslim Nagar Housing Society**, a centralized pumping station is proposed to handle peak wastewater flow and convey it to the disposal point efficiently.

9.1 Components of the Pumping Station

The pumping station comprises the following key components:

1. **Screening Chamber**: Removes large debris and solids to prevent pump damage.
2. **Wet Well**: Temporarily stores incoming wastewater before pumping.
3. **Pumps**: Lifts wastewater from the wet well to the desired elevation for onward flow.
4. **Control System**: Includes float switches and automated controls for pump operation.

9.2 Design Criteria

1. **Flow Capacity**:

- Peak Flow = **0.71 ft³/s** (as calculated in Hydraulic Design).
- A detention time of **10 minutes** is selected as per PHED Punjab standards.

2. **Wet Well Dimensions**:

Volume of wet well is determined using the formula:

$$V = Q \times T$$

Where:

- V: Wet well volume (ft³).
- Q: Peak inflow rate (0.35 ft³/s)
- T: Detention time (10 min=600 s)

$$V = 0.35 \times 600 = 210\, ft^3$$

Wet well cross-sectional area is designed as a **square**:

$$Area = \frac{Volume}{Depth}$$

Assuming a depth of **10 ft**:

$$Area = 210/10 = 21\, ft^2$$

Dimensions of wet well: **4.6 ft × 4.6 ft** (square).

3. **Pump Selection**:

   - **Centrifugal Submersible Pumps** are chosen for efficiency and ease of maintenance.
   - Required head (H) includes:

     - Static head: **15 ft** (elevation difference).
     - Friction head: **calculated using Hazen-Williams equation** for rising main.

   - Estimated total head: **20 ft**.
   - Required pump power (P) is calculated using:

$$P = \frac{\gamma \cdot Q \cdot H}{Efficiency}$$

   Where:

   - γ: Specific weight of water (62.4 lbs/ft³)
   - Q: Flow rate (0.35 ft³/s)
   - H: Total head (20 ft)
   - Efficiency: Assumed to be 65%

   Substituting values:

$$P = \frac{62.4 \cdot 0.35 \cdot 200}{0.65} = 668.31\,W \approx 9\,HP$$

   Pump power: **9 HP**.

4. **Standby Pumps**:

To ensure continuous operation, one standby pump is proposed.

9.3 Summary of Pumping Station Design

The proposed pumping station includes:

1. A **screening chamber** to remove debris and protect the pumps.
2. A **wet well** with a volume of **210 ft³**, designed for a detention time of 10 minutes.
3. Two **centrifugal submersible pumps** (1 operational, 1 standby) with a power requirement of **9 HP**.
4. Automated control systems for efficient operation.

This design ensures the reliable conveyance of wastewater to the disposal point under peak flow conditions.

10 Drawings of Wet Well and Screening Chamber

The drawings of Wet Well and Screening chamber are shown in Figure.

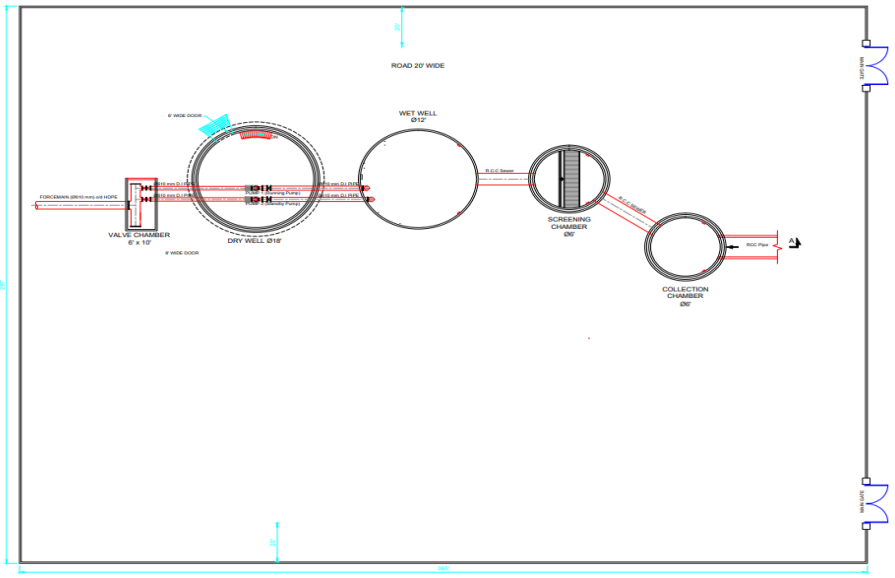


Figure 6. Sewer Pumping Station

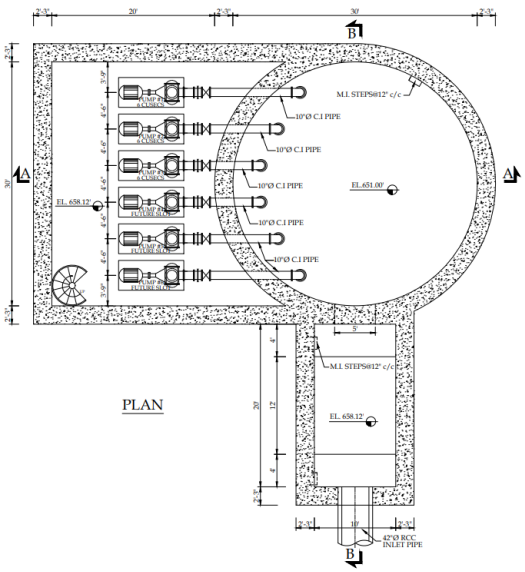


Figure 7. Sewer Pumping Station Layout Plan

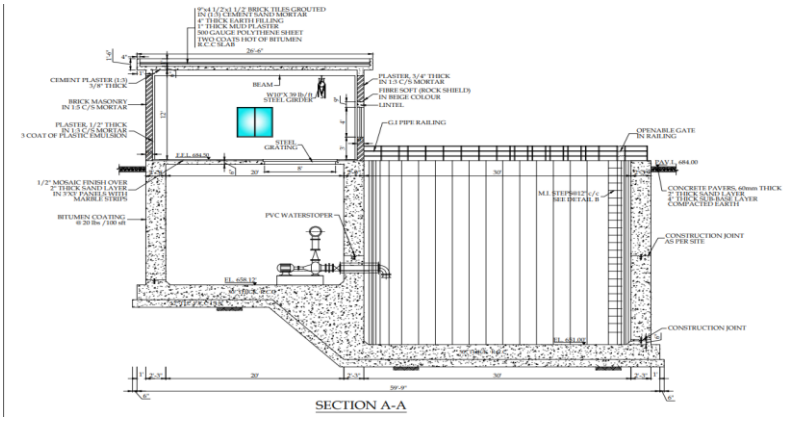


Figure 8. Sewer Pumping Station- Cross Section View with Structural Details Section A-A

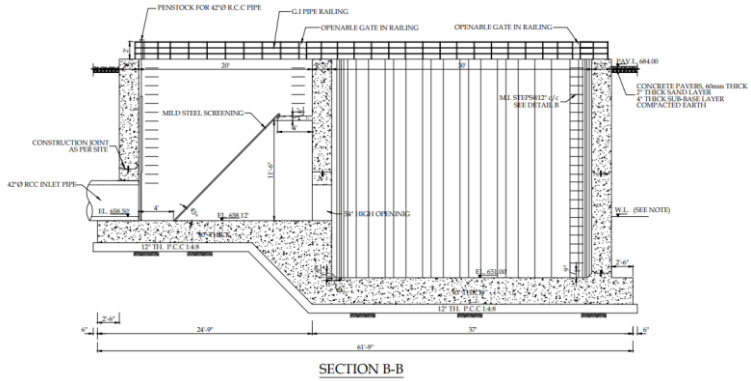


Figure 9. Sewer Pumping Station- Cross Section View with Structural Details Section B-B

11 Outfall Sewer

11.1 Introduction

The outfall sewer is a critical component of the sewerage system, responsible for transporting treated or untreated wastewater to the final disposal point, such as a natural water body or a treatment plant. The profile design of the outfall sewer must ensure:

1. **Adequate Conveyance Capacity**: To handle the peak sewage flow.
2. **Hydraulic Compliance**: Meeting velocity and slope requirements.
3. **Structural Integrity**: Resistance to external loads and environmental conditions.

11.2 Design Considerations for Outfall Sewer

1. **Hydraulic Capacity**:

    - The outfall sewer must accommodate the **peak flow of 16.98 ft³/s** (as calculated in the report).
    - A minimum velocity of 2.5 ft/s is maintained for self-cleansing, while the maximum velocity is capped at 10 ft/s to prevent pipe erosion.

2. **Material Selection**:

    - **Reinforced Cement Concrete (RCC)** pipes are chosen for their durability and resistance to corrosion and heavy loads.
    - Pipe joints will be watertight to prevent infiltration and exfiltration.



Figure 10. Profile layout

3. **Slope and Depth**:

    - The slope is calculated using Manning's equation to maintain self-cleansing velocity.
    - The minimum cover for the outfall sewer is **3 ft** to protect against traffic and earth loads.

4. **Alignment and Profile**:

    - The outfall sewer follows the natural terrain to minimize excavation.
    - Manholes are placed at junctions and changes in direction or slope.

11.3 Design Process Using SewerGEMS

The outfall sewer was designed using **SewerGEMS** to optimize the layout and hydraulic profile:

1. **Input Data**:

    - Peak flow: 16.98 ft^3/s.
    - Pipe material: RCC.
    - Manning's coefficient (nnn): 0.013.
    - Terrain data: Imported from GIS/CAD.

2. **Simulation**:

    - The hydraulic grade line (HGL) was analyzed to ensure gravity flow.
    - Pipe diameters were trial-assigned and adjusted to meet flow and velocity requirements.

The output of the simulation is:

- Pipe diameters: Ranged from 18 to 36 inches depending on flow requirements.
- Velocities: Maintained between 2.5–4 ft/s in most sections.
- Slopes: Varied to align with the terrain while maintaining self-cleansing velocity.

12 Capital Costing

The capital cost estimation of the sewerage system for **Muslim Nagar Housing Society** involves calculating the costs of major components: sewer lines, manholes, pumping station, and any associated works. These estimates are based on PHED standards and the Market Rate Schedule (MRS) .

12.1 Cost Components

1. **Sewer Lines**:

    - Includes earthwork excavation, RCC pipes, pipe laying, backfilling, and compaction.
    - Pipes of various diameters (9", 12", 15", 18") are included.
    - Calculated based on MRS 2021 rates for each diameter.

2. **Manholes**:

    - Includes excavation, brick masonry, plaster, and installation of covers.
    - Standard manhole dimensions as per PHED standards.
    - Different manhole depths (e.g., 3 ft, 6 ft) are considered.

3. **Pumping Station**:

    - Covers civil, mechanical, and electrical works.
    - Includes excavation, RCC construction, pump installation, and wiring.
    - Pump details include power requirements, number of units, and standby provision.

4. **Additional Works**:

    - Valves, flush tanks, and other special structures.
    - Contingencies and PRA charges as per standard practice.

12.1.1 Sewer Lines

For sewer lines, the cost includes excavation, pipe material, laying, jointing, testing, and backfilling. Using the MRS rates provided in your file:

Table 4. Sewer lines cost

Diameter (inches)	Unit (Rft)	Rate (Rs./Rft)	Total Length (Rft)	Total Cost (Rs.)
9" (225 mm)	Rft	350	1500	350×1500=525,000350 \times 1500 = 525,000350×1500=525,000
12" (310 mm)	Rft	400	1000	400×1000=400,000400 \times 1000 = 400,000400×1000=400,000
15" (380 mm)	Rft	600	800	600×800=480,000600 \times 800 = 480,000600×800=480,000
18" (450 mm)	Rft	800	500	800×500=400,000800 \times 500 = 400,000800×500=400,000
Total				**1,805,000**

12.1.2 Manholes

Manhole costs include excavation, brick masonry, plaster, and covers. Different depths are accounted for based on PHED standards and MRS data:

Table 5. Manhole cost

Depth (ft)	Unit (Each)	Rate (Rs./Each)	Quantity (Nos.)	Total Cost (Rs.)
3 ft	Each	15,000	10	15,000×10=150,000
6 ft	Each	25,000	15	25,000×15=375,000
Total				**525,000**

12.1.3 Pumping Station

The pumping station involves civil, mechanical, and electrical works. The MRS details provide accurate unit rates for these components:

Table 6. pumping station cost

Component	Details	Cost (Rs.)
Civil Works	Includes excavation, RCC, plaster, etc.	1,200,000
Mechanical Works	Pumps, wet wells, and piping installations	800,000
Electrical Works	Wiring, switchboards, and automation	400,000
Total Pumping Station		**2,400,000**

12.1.4 Additional Works

Includes valves, flush tanks, and miscellaneous items:

Table 7. Additional work cost

Description	Rate (Rs.)	Quantity	Total Cost (Rs.)
Valves (10 Nos.)	12,000	10	12,000×10=120,000
Flush Tanks	50,000	4	50,000×4=200,000
Miscellaneous Works	-	-	150,000
Total			**470,000**

12.1.5 Abstract of Costs

Table 8. Abstract of cost

Component	Cost (Rs.)
Sewer Lines	1,805,000
Manholes	525,000
Pumping Station	2,400,000
Additional Works	470,000
Subtotal	**5,200,000**
Add 3% Contingencies	156,000
Add 5% PRA Charges	260,000
Grand Total	**5,616,000**

13 Staffing Levels

The successful operation and maintenance of the sewerage system in **Muslim Nagar Housing Society** require a well-structured staffing plan. This plan is tailored to the system's complexity, which includes gravity sewer lines, manholes, and a pumping station. Ensuring adequate staffing levels is essential to:

1. Maintain uninterrupted system functionality.
2. Address routine and emergency maintenance needs.
3. Ensure compliance with PHED operational standards.

13.1 Staffing Requirements

The staffing plan is divided into operational and maintenance roles, based on the two main system components:

1. **Gravity Sewerage Network**:

    - Includes sewer lines, manholes, and flush tanks.
    - Requires periodic inspection, flushing, and cleaning.

2. **Pumping Station**:
    - Requires monitoring of pumps, wet wells, and electrical systems.

13.2 Key Staffing Roles and Responsibilities

1. **System Supervisor (1 Position)**:

    - Oversees the overall operation and maintenance of the sewerage system.
    - Ensures compliance with PHED guidelines and safety protocols.
    - Handles scheduling and reporting to management.

2. **Technicians (4 Positions)**:

    - Conduct routine inspections of sewer lines, manholes, and the pumping station.
    - Operate flushing equipment for sewer lines with velocities below self-cleansing thresholds.
    - Perform minor repairs and coordinate with the supervisor for major issues.

3. **Pump Operators (2 Positions)**:

    - Manage the pumping station, including the operation of centrifugal pumps.
    - Monitor wet well levels and control pump operations to match flow rates.
    - Conduct daily checks on mechanical and electrical components.

4. **Laborers (6 Positions)**:

    - Assist technicians in cleaning, repairing, and maintaining sewer lines and manholes.
    - Perform manual flushing and excavation as required.
    - Support pump operators during major maintenance tasks.

5. **Administrative Staff (2 Positions)**:

    - Manage documentation, including maintenance schedules, incident reports, and inventory.
    - Handle procurement of spare parts and consumables.

13.2.1 Staffing Requirements Table

Table 9. Staff requirement table

Role	Number of Staff	Responsibilities
System Supervisor	1	Overall management, compliance, and reporting.
Technicians	4	Inspection, maintenance, and minor repairs.
Pump Operators	2	Pump operation, monitoring, and maintenance.
Laborers	6	Cleaning, flushing, and excavation.
Administrative Staff	2	Documentation and inventory management.
Total Staff	**15**	

13.3 Justification

1. **Gravity Sewer Network**:

    - Regular flushing is necessary for sections with marginal velocities (1–2 ft/s) and critical velocities (<1 ft/s). Flushing tanks and manual methods require skilled technicians and laborers.
    - The extensive network demands systematic inspections to prevent blockages or surcharges.

2. **Pumping Station**:

    - Continuous operation requires two operators (one per shift) to ensure reliability.
    - Additional laborers assist in wet well cleaning and emergency repairs.

3. **Administrative Efficiency**:

    - Proper documentation ensures smooth operation and facilitates audits.
    - Effective inventory management minimizes downtime.

13.4 Cost Implications

The estimated monthly salary for each role, based on market standards:

Table 10. Cost implication

Role	Monthly Salary (Rs.)	Number of Staff	Total Cost (Rs./Month)
System Supervisor	50,000	1	50,000
Technicians	40,000	4	160,000
Pump Operators	35,000	2	70,000
Laborers	25,000	6	150,000
Administrative Staff	30,000	2	60,000
Total Monthly Cost		**15**	**490,000**

Annual staffing cost: 490000 * 12 = **Rs. 5,880,000 (approximately Rs. 5.88 million)**.

14 Operation and Maintenance (O&M) Costs

Operation and Maintenance (O&M) costs are pivotal for ensuring the long-term sustainability of the sewerage system. These costs involve expenses for labor, energy, maintenance, consumables, and contingencies. Each element plays a critical role in keeping the system efficient and compliant with PHED standards. A balanced O&M plan not only addresses routine requirements but also prepares for emergencies and unforeseen repairs.

The following breakdown provides a detailed look into the annual O&M costs for the sewerage system at **Muslim Nagar Housing Society**, keeping in mind the complexity of the network and associated pumping station.

14.1 Components of O&M Costs

1. **Labor Costs**:

   A well-trained workforce is vital to ensure uninterrupted operations. The staffing plan already specifies roles like supervisors, technicians, pump operators, and laborers. These personnel will handle everything from routine inspections and repairs to pump monitoring and cleaning operations.

2. **Energy Costs**:

   Since the system includes a pumping station, energy consumption forms a significant part of the budget. The cost is calculated based on the pump’s power requirements, operational hours, and local electricity tariffs.

3. **Routine Maintenance**:

   Sewer lines, manholes, and flushing tanks require periodic cleaning and inspections to prevent blockages and overflows. This also includes replacing worn-out parts like valves and rubber rings.

4. **Consumables**:

   Items like lubricants, gaskets, and small spare parts are necessary for the smooth functioning of pumps and other components.

5. **Contingencies**:

   A contingency fund is essential to cover unexpected repairs or emergencies, ensuring there is no disruption in operations.

14.2 Detailed Cost Calculation

14.2.1 Labor Costs

Labor costs form the backbone of the O&M budget. Based on the staffing plan:

System Supervisor: 1 person at Rs. 50,000/month.

Technicians: 4 persons at Rs. 40,000/month each.

Pump Operators: 2 persons at Rs. 35,000/month each.

Laborers: 6 persons at Rs. 25,000/month each.

Administrative Staff: 2 persons at Rs. 30,000/month each.

$$\begin{aligned} &Monthly\ Labor\ Cost \\ &\quad = (50{,}000) + (4 \times 40{,}000) + (2 \times 35{,}000) + (6 \times 25{,}000) + (2 \times 30{,}000) \\ &\quad = 490{,}000\ Rs \end{aligned}$$

$$Annual\ Labor\ Cost = 490{,}000 \times 12 = 5{,}880{,}000\ Rs$$

14.2.2 Energy Costs

The pumping station's energy consumption is calculated based on a **1 HP pump** operating 24/7 (with a standby pump for emergencies).

- **Power Conversion**: 1 HP = 0.746 kW.
- **Monthly Energy Usage**: $0.746\ kW \times 24\ hrs/day \times 30\ days = 537.12\ kWh$
- **Monthly Energy Cost** (at Rs. 20/kWh):

$$537.12 \times 20 = 10{,}742\ Rs.$$

Annual Energy Cost: $10{,}742 \times 12 = 128{,}904\ Rs.$

14.2.3 Routine Maintenance

Maintenance ensures the system functions effectively. Costs include:

- **Sewer Cleaning and Manhole Maintenance**: Rs. 300,000/year.
- **Replacement Parts**: Rs. 150,000/year.
- **Flushing Tank Operations**: Rs. 50,000/year.
- **Total Maintenance Cost**:

$$300{,}000 + 150{,}000 + 50{,}000 = 500{,}000\,Rs.$$

14.2.4 Consumables

Consumables like lubricants and spare parts are essential for routine operations. Allocations are:

- **Lubricants**: Rs. 20,000/year.
- **Miscellaneous Items (e.g., gaskets, rubber rings)**: Rs. 30,000/year.
- **Total Consumables Cost**: Rs. 50,000/year.

14.2.5 Contingencies

To prepare for unexpected failures or emergencies, **10% of the total annual costs (excluding contingencies)** is allocated:

- Contingencies=$10\% \times (5{,}880{,}000 + 128{,}904 + 500{,}000 + 50{,}000) = 656{,}890\,Rs.$

14.2.6 O&M Cost Summary

Table 11. O&M cost summary

Component	Annual Cost (Rs.)
Labor Costs	5,880,000
Energy Costs	128,904
Routine Maintenance	500,000
Consumables	50,000
Contingencies	656,890
Total O&M Costs	**7,214,894**

15 Per Household Sewer Bill

Household bill is calculated to distribute the annual **Operation & Maintenance (O&M)** costs among the residents of **Muslim Nagar Housing Society** equitably. This calculation ensures that the sewerage system remains financially sustainable while being affordable for the residents. The sewer bill covers labor costs, energy expenses, routine maintenance, consumables, and contingencies.

15.1 Data for calculation:

1. **Total Annual O&M Costs**: Rs. 7,214,894
2. **Number of Households**: **350 households** (based on the project's description).
3. **Monthly Distribution**: The O&M costs are divided among households and spread over 12 months.

15.2 Calculation

1. **Annual Cost Per Household**:

$$Annual\ Cost\ Per\ Household = Total\ Annual\ O\&M\ Cost / Total\ Number\ of\ Households$$

Substituting the values:

$$Annual\ Cost\ Per\ Household = 7{,}214{,}894416 = 17{,}344\ Rs$$

2. **Monthly Sewer Bill Per Household**:

$$Monthly\ Sewer\ Bill = \frac{Annual\ Cost\ Per\ Household}{12}$$

Substituting the values:

$$Monthly\ Sewer\ Bill = \frac{17{,}344}{12} \approx 1{,}445\ Rs$$

15.3 Justifications

The system ensures fair cost distribution by having each household pay an equal share of the O&M expenses, which contributes to its long-term sustainability. A monthly bill of Rs. 1,445 is considered affordable for middle and upper-middle-class residents, offering sufficient funds for maintenance, energy, and contingency reserves. Additionally, a portion of the contingency funds ensures that households are not burdened with extra costs during emergencies.

15.4 Recommendations

To ensure smooth operation, it is recommended to implement a reliable billing system for timely collection and provide digital payment options for improved convenience. Future adjustments to the O&M costs should be made annually, taking into account inflation, changes in energy tariffs, or system upgrades. It is also essential to maintain transparency by providing households with an annual report that details how the funds are spent, fostering trust and accountability.

16 References

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17 Annexure

Calculation Detailed Summary

<General>			
Label	Base Calculation Options	Output Increment	0.050 hours
Calculation Time Step	0.025 hours	Duration	24.000 hours

Implicit Engine			
Virtual Flow Depth	0.040 ft	NR Weighting Coefficient	0.700
Y Iteration Tolerance	0.03 ft	Relaxation Weighting Coefficient	0.600
LPI Coefficient	1.000	Computation Distance	50.00 ft

Inlets			
Neglect Side Flow?	False	Active Components for Combination Inlets In Sag	Grate and Curb
Neglect Gutter Cross Slope For Side Flow?	False	Active Components for Combination Inlets on Grade	Grate and Curb

Catchment Calculation Summary

Label	Runoff Method	Loss Method	Total Rainfall Depth (in)	Area (User Defined) (acres)	Volume (Total Runoff) (gal)	Flow (Maximum) (cfs)	Time (Maximum Flow) (hours)

General Calculation Summary

Label	Element Type	Branch	Time (Maximum Flow) (hours)	Flow (Maximum) (cfs)	Velocity (Maximum Calculated) (ft/s)
MH-1	Manhole	15	---	---	---
MH-2	Manhole	16	---	---	---
MH-3	Manhole	15	---	---	---
MH-4	Manhole	17	---	---	---
MH-5	Manhole	15	---	---	---
MH-6	Manhole	18	---	---	---
MH-7	Manhole	18	---	---	---
MH-8	Manhole	15	---	---	---
MH-9	Manhole	11	---	---	---
MH-10	Manhole	12	---	---	---
MH-11	Manhole	12	---	---	---
MH-12	Manhole	11	---	---	---
MH-13	Manhole	13	---	---	---
MH-14	Manhole	13	---	---	---
MH-15	Manhole	13	---	---	---
MH-16	Manhole	11	---	---	---
MH-17	Manhole	14	---	---	---
MH-18	Manhole	14	---	---	---
MH-19	Manhole	11	---	---	---

General Calculation Summary

Label	Element Type	Branch	Time (Maximum Flow) (hours)	Flow (Maximum) (cfs)	Velocity (Maximum Calculated) (ft/s)
MH-20	Manhole	8	---	---	---
MH-21	Manhole	8	---	---	---
MH-22	Manhole	9	---	---	---
MH-23	Manhole	8	---	---	---
MH-24	Manhole	10	---	---	---
MH-25	Manhole	8	---	---	---
MH-26	Manhole	5	---	---	---
MH-27	Manhole	7	---	---	---
MH-28	Manhole	6	---	---	---
MH-29	Manhole	6	---	---	---
MH-30	Manhole	6	---	---	---
MH-31	Manhole	5	---	---	---
MH-32	Manhole	5	---	---	---
MH-33	Manhole	5	---	---	---
MH-34	Manhole	5	---	---	---
MH-35	Manhole	5	---	---	---
MH-36	Manhole	5	---	---	---
MH-37	Manhole	19	---	---	---
MH-38	Manhole	5	---	---	---
MH-39	Manhole	1	---	---	---
MH-40	Manhole	2	---	---	---
MH-41	Manhole	2	---	---	---
MH-42	Manhole	2	---	---	---
MH-43	Manhole	4	---	---	---
MH-44	Manhole	4	---	---	---
MH-45	Manhole	4	---	---	---
MH-46	Manhole	3	---	---	---
MH-47	Manhole	3	---	---	---
MH-48	Manhole	3	---	---	---
MH-49	Manhole	1	---	---	---
MH-50	Manhole	1	---	---	---
MH-51	Manhole	1	---	---	---
CO-1	Conduit	15	0.400	0.05	1.22
CO-2	Conduit	15	0.200	0.13	1.84
CO-3	Conduit	15	0.150	0.15	1.98
CO-4	Conduit	15	0.150	0.30	2.94
CO-5	Conduit	5	0.200	1.87	4.02
CO-6	Conduit	5	0.200	1.98	4.18
CO-8	Conduit	4	0.000	0.07	1.17
CO-9	Conduit	4	0.450	0.13	1.81
CO-10	Conduit	4	0.050	0.17	2.07
CO-11	Conduit	3	0.150	0.39	3.26
CO-12	Conduit	1	0.200	0.64	3.52
CO-13	Conduit	18	0.000	0.04	1.04
CO-14	Conduit	18	9.000	0.09	1.53
CO-15	Conduit	14	0.100	0.04	1.11
CO-16	Conduit	14	0.000	0.16	1.83

General Calculation Summary

Label	Element Type	Branch	Time (Maximum Flow) (hours)	Flow (Maximum) (cfs)	Velocity (Maximum Calculated) (ft/s)	Hydraulic Grade (Maximum) (ft)
CO-17	Conduit	11	0.750	0.55	3.59	96.20
CO-19	Conduit	11	9.000	0.02	0.84	96.19
CO-20	Conduit	11	0.600	0.17	2.10	94.95
CO-21	Conduit	11	0.750	0.33	2.90	96.14
CO-22	Conduit	12	0.000	0.04	0.96	94.27
CO-23	Conduit	12	0.000	0.08	1.33	96.20
CO-24	Conduit	13	0.000	0.02	0.67	95.66
CO-25	Conduit	13	0.500	0.09	1.53	92.66
CO-26	Conduit	13	0.000	0.12	1.42	
CO-27	Conduit	8	9.000	0.07	1.39	
CO-28	Conduit	8	0.100	0.11	1.68	
CO-29	Conduit	8	0.300	0.28	2.79	
CO-30	Conduit	8	0.000	1.12	6.74	
CO-31	Conduit	5	0.250	0.83	4.12	
CO-32	Conduit	5	0.100	1.49	4.21	
CO-33	Conduit	6	9.000	0.06	1.31	
CO-34	Conduit	6	9.000	0.12	1.76	
CO-35	Conduit	6	0.550	0.22	2.39	
CO-36	Conduit	5	9.000	0.30	2.93	
CO-37	Conduit	5	0.250	0.37	3.15	
CO-38	Conduit	5	0.050	0.43	2.99	
CO-39	Conduit	2	0.450	0.02	0.84	
CO-40	Conduit	2	0.000	0.06	1.25	
CO-41	Conduit	2	0.000	0.12	1.73	
CO-42	Conduit	3	0.000	0.07	1.14	
CO-43	Conduit	3	0.450	0.14	1.89	
CO-44	Conduit	1	0.350	2.71	4.10	
CO-45	Conduit	16	0.500	0.04	1.10	
CO-46	Conduit	17	0.200	0.01	0.74	
CO-47	Conduit	19	0.100	0.06	1.30	
CO-48	Conduit	1	0.450	0.06	1.28	
CO-49	Conduit	10	0.000	0.03	0.72	
CO-50	Conduit	9	0.300	0.07	1.39	
CO-51	Conduit	5	0.400	0.06	1.31	
CO-52	Conduit	7	0.000	0.04	1.10	
CO-53	Conduit	1	0.150	2.77	4.53	

General Calculation Summary

Hydraulic Grade (Maximum) (ft)
96.15
96.20
95.41
94.98
96.15
94.93
94.08
93.31
96.19
94.47
92.37
96.21
94.92
96.21
94.25
96.18
93.46
96.21
96.19
96.21
95.05
94.37
94.98
92.53
91.31
87.89
85.72
84.16
96.21
83.39
96.21
96.15
94.21
92.35
96.21
95.57
94.84
96.21
94.31
93.62
94.78
90.35
81.82
96.20
94.95
94.27
92.66
84.06

General Calculation Summary

Hydraulic Grade (Maximum) (ft)
83.39
96.21
95.57
94.84
92.90
91.99
96.20
95.66
96.19
94.47
92.37
96.15
94.98
93.31
96.20
95.41
96.15
94.93
94.08
96.21
94.92
94.25
93.46
87.89
85.65
96.21
95.05
94.37
93.71
92.53
91.31
96.15
94.21
92.35
96.21
94.31
82.90
96.19
96.14
96.21
96.21
96.18
96.21
96.21
96.19
81.81

Node Calculation Summary

Node Calculation Summary

Label	Element Type	Branch	Time to Maximum Inflow (hours)	Flow (Total In Maximum) (cfs)	Time To Maximum Inlet Flow (hours)
MH-1	Manhole	15	0.000	0.05	---
MH-2	Manhole	16	0.000	0.04	---
MH-3	Manhole	15	0.100	0.13	---
MH-4	Manhole	17	0.000	0.01	---
MH-5	Manhole	15	0.150	0.15	---
MH-6	Manhole	18	0.000	0.04	---
MH-7	Manhole	18	0.400	0.09	---
MH-8	Manhole	15	0.100	0.30	---
MH-9	Manhole	11	0.000	0.02	---
MH-10	Manhole	12	0.000	0.04	---
MH-11	Manhole	12	0.050	0.08	---
MH-12	Manhole	11	0.000	0.18	---
MH-13	Manhole	13	0.000	0.02	---
MH-14	Manhole	13	0.100	0.09	---
MH-15	Manhole	13	0.000	0.12	---
MH-16	Manhole	11	0.150	0.33	---
MH-17	Manhole	14	0.000	0.04	---
MH-18	Manhole	14	0.000	0.16	---
MH-19	Manhole	11	9.050	0.55	---
MH-20	Manhole	8	0.000	0.07	---
MH-21	Manhole	8	9.000	0.11	---
MH-22	Manhole	9	0.000	0.07	---
MH-23	Manhole	8	0.100	0.28	---
MH-24	Manhole	10	0.000	0.03	---
MH-25	Manhole	8	0.200	0.37	---
MH-26	Manhole	5	0.000	0.06	---
MH-27	Manhole	7	0.000	0.04	---
MH-28	Manhole	6	0.000	0.06	---
MH-29	Manhole	6	9.000	0.12	---
MH-30	Manhole	6	0.100	0.22	---
MH-31	Manhole	5	0.400	0.30	---
MH-32	Manhole	5	9.000	0.37	---
MH-33	Manhole	5	0.500	0.42	---
MH-34	Manhole	5	0.000	1.37	---
MH-35	Manhole	5	0.650	1.48	---
MH-36	Manhole	5	0.100	1.88	---
MH-37	Manhole	19	0.000	0.06	---
MH-38	Manhole	5	0.200	1.98	---
MH-39	Manhole	1	0.000	0.06	---
MH-40	Manhole	2	0.000	0.02	---
MH-41	Manhole	2	0.050	0.06	---
MH-42	Manhole	2	0.050	0.12	---
MH-43	Manhole	4	0.000	0.07	---
MH-44	Manhole	4	0.000	0.13	---
MH-45	Manhole	4	0.250	0.17	---
MH-46	Manhole	3	0.000	0.06	---
MH-47	Manhole	3	0.050	0.14	---

Node Calculation Summary

Label	Element Type	Branch	Time to Maximum Inflow (hours)	Flow (Total In Maximum) (cfs)	Time To Maximum Inlet Flow (hours)	Flow (Surface Maximum) (cfs)	Time To Maximum Captured Flow (hours)	Flow (Captured Maximum) (cfs)	Time to Maximum Overflow (hours)	Flow (Overflow Maximum) (cfs)
MH-48	Manhole	3	0.000	0.40	---	---	---	---	0.000	0.00
MH-49	Manhole	1	1.150	0.64	---	---	---	---	0.000	0.00
MH-50	Manhole	1	0.200	2.71	---	---	---	---	0.000	0.00
MH-51	Manhole	1	0.150	2.76	---	---	---	---	0.000	0.00
O-1	Outfall	1	0.150	2.77	---	---	---	---	0.000	0.00

Node Calculation Summary

Flow (Surface Maximum) (cfs)	Time To Maximum Captured Flow (hours)	Flow (Captured Maximum) (cfs)	Time to Maximum Overflow (hours)	Flow (Overflow Maximum) (cfs)
---	---	---	0.000	0.00
---	---	---	0.000	0.00
---	---	---	0.000	0.00
---	---	---	0.000	0.00
---	---	---	0.000	0.00
---	---	---	0.000	0.00
---	---	---	0.000	0.00
---	---	---	0.000	0.00
---	---	---	0.000	0.00
---	---	---	0.000	0.00
---	---	---	0.000	0.00
---	---	---	0.000	0.00
---	---	---	0.000	0.00
---	---	---	0.000	0.00
---	---	---	0.000	0.00
---	---	---	---	---

Gutter Calculation Summary

Label	Gutter Shape	Flow (Maximum) (cfs)	Time (Maximum Flow) (hours)	Velocity (Maximum Calculated) (ft/s)

Calculation Executive Summary

Scenario	
Label	Base

Storm Event			
Label	Base Rainfall Runoff	Return Event	(N/A) years
Global Storm Event	<None>		

Calculation Executive Summary			
Total Inflow Volume	1,779,256.1 gal	Total System Volume Change	3,619.8 gal
Total System Outflow Volume	1,776,800.3 gal	Continuity Error	0.1 %
Total System Overflow Volume	0.0 gal	Total Pond Infiltration Volume	0.0 gal
Total Gutter Volume Change	(N/A) gal	Total N-R Iterations	18691